DirtBike Maintence System

IoT Predictive Maintenance for Dirt Bikes

CSC 494-001
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1. Problem & Importance

Problem

Dirt bikes face dust, high RPMs, crashes → neglected maintenance causes 80–90% of engine "blow-ups"

- Low oil → seizure, scored bearings
- Low coolant → overheating, warped heads
- Clogged air filter → power loss, abrasive wear

Why important?

Prevents costly rebuilds, mid-ride failures, enhances safety on

2. Project Topic

Software focus

Build IoT app on ESP32 hardware:

- Real-time sensor data processing
- Secure cloud transmission
- Mobile dashboard + alerts

Hardware used

- Sensors: ultrasonic (levels), differential pressure (air box)
- ESP32
- Bluetooth

3. Two Topics to Learn with AI

Software

Secure IoT comms & cloud integration

- Build dashboard
- Alert logic + buffering

Hardware

ESP32 sensor interfacing & calibration

- Ultrasonic & differential pressure
- 12V bike power regulation

4. Time Usage – Two Iterations

Iteration 1 (Sprint 1)
Learn → build

- Sensor reading/calibration
- Basic dashboard + thresholds

Iteration 2 (Sprint 2)
Full features

- Alerts (push/email)
- Trends, local buffering
- Vibration/dust testing

5. Final Goal (End of Semester)

- **Working prototype** mounted & demonstrated
- **Research paper**: show findings, bugs, fixes, and w
- **Portfolio piece**: GitHub repo, code, schematics, demo video
- **Future extension** potential: vibration sensor

Questions?